Supplementary material

Supplementary File 1. PRISMA 2009 checklist[21]

Section/topic	Checklist item	Reported on page #
Title		
1. Title	Identify the report as a systematic review, meta-analysis, or both.	1
Abstract		
2. Structured summary	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	2
Introduction		
3. Rationale	Describe the rationale for the review in the context of what is already known.	4
4. Objectives	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
Methods		
5. Protocol and registration	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
6. Eligibility criteria	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
7. Information sources	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
8. Search	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Suppl File 2
9. Study selection	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
10. Data collection process	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
11. Data items	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Suppl File 3
12. Risk of bias in individual studies	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6

13. Summary measures	State the principal summary measures (e.g., risk ratio, difference in means).	6
14. Synthesis of results	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I2) for each meta-analysis.	6
15. Risk of bias across studies	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
16. Additional analyses	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
Results		
17. Study selection	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8 and Fig 2
18. Study characteristics	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8 and Table 2
19. Risk of bias within studies	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8 and Table 2
20. Results of individual studies	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8
21. Synthesis of results	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	22-26
22. Risk of bias across studies	Present results of any assessment of risk of bias across studies (see Item 15).	8 and Table 2
23. Additional analysis	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
Discussion		
24. Summary of evidence	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	32-33
25. Limitations	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	33
26. Conclusions	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	33
Funding		
27. Funding	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	34

Supplementary File 2. Search terms

OVID (MEDLINE, Global Health, PsychINFO)

(Angola.mp. OR Armenia.mp. OR Bangladesh.mp. OR Bhutan.mp. OR Bolivia.mp. OR Cape Verde.mp.OR Cambodia.mp. OR Cameroon.mp. OR Congo.mp. OR Ivory Coast.mp. OR Djibouti.mp. OR Egypt.mp OR El Salvador.mp. OR Georgia.mp. OR Ghana.mp. OR Guatemala.mp. OR Honduras.mp. OR India.mp. OR Indonesia.mp. OR Jordan.mp. OR Kenya.mp. OR Kiribati.mp. OR kosovo.mp OR Kyrgyz Republic.mp. OR Laos.mp. OR Lesotho.mp. OR Mauritania.mp. OR Micronesia.mp. OR Moldova.mp. OR Mongolia.mp. OR Morocco.mp. OR Myanmar.mp. OR Nicaragua.mp. OR Nigeria.mp. OR Pakistan.mp. OR Papua New Guinea.mp. OR Philippines.mp. OR (Sao Tome and Principe).mp. OR Solomon Islands.mp. OR Sri Lanka.mp. OR Sudan.mp. OR Swaziland.mp. OR Syria.mp. OR Tajikistan.mp. OR Timor.mp. OR Tunisa.mp. OR Ukraine.mp. OR Uzebekistan.mp. OR Vanuatu.mp. OR Vietnam.mp. OR Palestine.mp. OR Yemen.mp. OR Zambia.mp. OR Afghanistan.mp. OR Benin.mp. OR Burkina Faso.mp. OR Burundi.mp. OR Central African Republic.mp. OR Chad.mp. OR Comoros.mp. OR Eritrea.mp. OR Ethiopia.mp. OR Gambia.mp. OR Guinea.mp. OR Haiti.mp. OR North Korea.mp. OR Liberia.mp. OR Madagascar.mp. OR Malawi.mp. OR Mali.mp. OR Mozambique.mp. OR Nepal.mp. OR Niger.mp. OR Rwanda.mp. OR Senegal.mp. OR Sierra Leone.mp. OR Somalia.mp. OR Tanzania.mp. OR Togo.mp. OR Uganda.mp. OR Zimbabwe.mp.; Annotation: key word all countries; Annotation: expand all countries+key word all countries) **AND** (Cancer.mp. OR Neoplasm.mp. OR Tumour.mp. OR Tumor.mp. OR Neoplasm.mp. OR Melanoma.mp. OR Carcinoma.mp. OR Sarcoma.mp. OR Leukaemia.mp. OR Malignancy.mp. OR Metastasis.mp. OR Adenocarcinoma.mp. OR Lymphoma.mp.) **AND** (Help seeking.mp. OR Health seeking.mp. OR Seeking Help.mp. OR Help-seeking.mp. OR Health-seeking.mp. OR Barrier*.mp. OR Delay*.mp. OR Early detection.mp. OR Late detection.mp. OR Late presentation.mp. OR Early presentation.mp. OR Delayed presentation.mp. OR Delayed detection.mp. OR Delaying presentation.mp. OR Presentation Delay.mp. OR Time to present*.mp. OR First present*.mp. OR Initial Present*.mp. OR Barrier to present*.mp. OR Early diagnosis.mp. OR Late diagnosis.mp. OR Delayed diagnosis.mp. OR Delaying diagnosis.mp. OR Diagnosis delay.mp. OR Barrier to diagnosis.mp. OR Time to diagnosis.mp. OR Time to treatment.mp. OR Barrier to treat*.mp. OR Treatment delay.mp. OR Delaying treatment.mp. OR Self-assessment.mp. OR Support seeking.mp. OR Seeking support.mp.) **AND** (Aware*.mp. OR Understand*.mp. OR Knowledge.mp. OR Recogni*.mp. OR Experience*.mp. OR Belie*.mp. OR Attitude*.mp. OR Thought*.mp. OR View*.mp. OR Perce*.mp. OR Perspective*.mp. OR Feel.mp. OR Feel*.mp. OR Emotion*.mp. OR Worr*.mp. OR Concern*.mp. OR Anxiet*.mp. OR Hope*.mp. OR Rural*.mp. OR Resource*.mp. OR Barrier*.mp. OR Psychosocial.mp. OR sociocultural.mp. OR Psycho-social.mp. OR Socio-cultural.mp.) **AND** (Questionnaire.mp. OR Survey.mp. OR Interview.mp. OR Focus Group.mp. OR Observ*.mp. OR Study.mp. OR Chart.mp. OR Qualitative.mp. OR Quantitative.mp. OR Mixed Method*.mp.)

Cochrane library

(Angola OR Armenia OR Bangladesh OR Bhutan OR Bolivia OR (Cape Verde) OR Cambodia OR Cameroon OR Congo OR (Ivory Coast) OR Djibouti. OR Egypt OR (El Salvador) OR Georgia

OR Ghana OR Guatemala OR Honduras OR India OR Indonesia OR Jordan OR Kenya OR Kiribati OR Kosovo OR (Kyrgyz Republic) OR Laos OR Lesotho OR Mauritania OR Micronesia OR Moldova OR Mongolia OR Morocco OR Myanmar OR Nicaragua OR Nigeria OR Pakistan OR (Papua New Guinea) OR Philippines OR (Sao Tome and Principe) OR Solomon Islands OR Sri Lanka OR Sudan OR Swaziland OR Syria OR Tajikistan OR Timor OR Tunisa OR Ukraine OR Uzebekistan OR Vanuatu OR Vietnam OR Palestine OR Yemen OR Zambia OR Afghanistan OR Benin OR Burkina Faso OR Burundi OR (Central African Republic) OR Chad OR Comoros OR Eritrea OR Ethiopia OR Gambia OR Guinea OR Haiti OR (North Korea) OR Liberia OR Madagascar OR Malawi OR Mali OR Mozambique OR Nepal OR Niger OR Rwanda OR Senegal OR (Sierra Leone) OR Somalia OR Tanzania OR Togo OR Uganda OR Zimbabwe) (NB word variation searched) **AND** (Cancer OR Neoplasm OR Tumour OR Tumor OR Neoplasm OR Melanoma OR Carcinoma OR Sarcoma OR Leukaemia OR Malignancy OR Metastasis OR Adenocarcinoma OR Lymphoma) AND (Barrier* or (Help seeking) OR (Health seeking) OR (Seeking Help) OR (Help-seeking) OR (Health-seeking) OR (Early detection) OR (Late detection) OR (Late presentation) OR (Early presentation) OR (Delayed presentation) OR (Delayed detection) OR (Delaying presentation) OR (Presentation Delay) OR (Time to present*) OR (First present*) OR (Initial Present*) OR (Barrier to present*) OR (Early diagnosis) OR (Late diagnosis) OR (Delayed diagnosis) OR (Delaying diagnosis) OR (Diagnosis delay) OR (Barrier to diagnosis) OR (Time to diagnosiS) OR (Time to treatment) OR (Barrier to treat*) OR (Treatment delay) OR (Delaying treatment) OR (Self-assessment) OR (Support seeking) OR (Seeking support) or Delay*) **AND** (Aware* OR Understand* OR Knowledge OR Recogni* OR Experience* OR Belie* OR Attitude* OR Thought* OR View*OR Perce* OR Perspective* OR Feel OR Feel* OR Emotion* OR Worr* OR Concern* OR Anxiet* OR Hope* OR Rural* OR Resource* OR Barrier* OR sociocultural OR (Psycho-social) OR (Socio-cultural) OR Psychosocial) **AND** (Questionnaire OR Survey OR Interview OR (Focus Group) OR Observ* OR Study OR Chart OR Qualitative OR Quantitative OR Mixed Method*)

Supplementary File 3. Extracted variables

Main heading	Variables extracted
Study characteristics	Lead author; year of publication; last year of data collection (where data collection year was not reported the last data collection year was calculated by year of publication minus two years); tumour site

Country characteristics	Country where the study was conducted; LMIC status in period of data collection (); GDP per capita in period of data collection
Design characteristics	Psychological model used; study design (observed/hypothetical; qualitative/quantitative/mixed-methods); data collection method; how the survey/topic guide were developed
Sample characteristics	Sample demographics (number of participants; gender; age; marital status; educational attainment; employment; income; area of residence; socioeconomic status; size of tumour at diagnosis; health insurance); method of recruitment
Help-seeking behaviour	Time period; mean/median time in the patient, diagnostic or treatment intervals; how 'delay' was defined and proportion of participants who 'delayed' in the patient, diagnostic or treatment intervals; proportion of participant who would/did seek help in the patient, diagnostic or treatment intervals.
Psychosocial variables	Symptom knowledge; symptom interpretation; social influences; use of TCAM medicines; cancer beliefs; competing priorities; expectations/beliefs about care seeking; availability and access to care; financial barriers; understanding/navigating the health system; cultural barriers; cancer treatment beliefs; risk factor awareness